

A One-Parameter, Four-Channel Consistency Test for EFC Background Coupling ($\beta \cdot T(a)$)

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Abstract

We test a one-parameter modification of the Friedmann equation in Energy-Flow Cosmology (EFC), where the coupling is routed through the background expansion: $H^2(a) = H_0^2[\Omega_m a^{-3} + \Omega_\Lambda(1 + \beta \cdot T(a))]$, with a fixed transition function $T(a)$ that is suppressed when $\Omega_m \rightarrow 1$. We show that (i) Poisson-equation modification ($\mu > 1$) produces growth enhancement and is disfavoured by BOSS DR12 redshift-space distortions, while (ii) background coupling ($\beta > 0$) produces growth suppression via enhanced Hubble friction and is consistent with data.

For $\beta \approx 0.08$, the combined BAO + RSD data show a mild preference ($\Delta\chi^2 = -3.89$ for one additional parameter on 9 data points), without introducing regime-specific patches. The same parameter is independently tested against CMB lensing (ACT+SPT+Planck, 2025) and Type Ia supernovae (Pantheon+ range), both consistent within 1σ . The model produces a null effect at $z = 3$ (Ly α P1D) and a GR-recovery envelope at recombination with margins exceeding 10^8 .

The result is a testable architectural constraint: EFC behaves as a dynamical dark-energy-like background effect, not as classical modified gravity in the Poisson term. Four independent cosmological channels, one parameter, zero kills.

Scope of this paper

What this is: A consistency and direction test for EFC's background coupling channel. One parameter (β), one transition function $T(a)$, four independent cosmological probes. The test verifies regime consistency within a standard FRW calibration.

What this is not: A full MCMC fit, a modified Boltzmann calculation, or a background-free EFC cosmology. All observational quantities are computed using Λ CDM-calibrated distances and growth factors.

1. Introduction

A central challenge for any modification of General Relativity is regime consistency: a late-time modification must recover standard gravity during matter domination, at recombination, and at Big Bang Nucleosynthesis. Most approaches achieve this through screening mechanisms, scale-dependent operators, or piecewise transition functions.

Energy-Flow Cosmology (EFC) takes a different approach. The modification strength is tied to the cosmic energy composition through a transition function $T(a)$ that scales with the non-matter fraction $(1 - \Omega_m(z))$. When matter dominates, $T \rightarrow 0$ and GR is recovered automatically. This is not a screening mechanism — it is a constitutive law.

A separate structural question is *where* the coupling enters the field equations. Two channels are possible: modification of the Poisson equation (affecting gravitational potentials directly) or modification of the Friedmann equation (affecting the expansion rate). These have opposite effects on structure growth, making the distinction empirically testable.

This paper presents the first joint test of EFC’s background coupling against four independent cosmological probes: BAO distances, redshift-space distortions, CMB lensing, and Type Ia supernovae.

2. The Modified Friedmann Equation

2.1 Coupling Architecture

The EFC background modification takes the form

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_\Lambda (1 + \beta \cdot T(a))]$$

where β is a dimensionless coupling amplitude and $T(a)$ is the standard EFC transition function, defined from the distance-coupling kernel $g_D(z) = [1 - \Omega_m(z)] / (1+z) \times \ln(1+z)$, normalised to peak at unity. At high redshift, $T \rightarrow 0$ and Λ CDM is recovered exactly.

2.2 Why Not the Poisson Channel?

The alternative coupling — modifying the Poisson equation as $k^2\Phi = -4\pi G a^2 \mu(a) \rho_m \delta$ with $\mu > 1$ — enhances gravitational attraction and therefore *increases* structure growth. This is the wrong direction relative to the observed S_8 tension.

Channel	χ^2 (RSD)	$\sigma_{8,0}$	Direction	vs S_8	Status
Poisson ($\alpha = 0.34$)	18.2	0.879	Enhanced	Wrong	Excluded
Background ($\beta = 0.08$)	1.09	0.805	Suppressed	Correct	Favoured
Λ CDM ($\alpha = \beta = 0$)	1.34	0.811	—	—	Baseline

Table 1. Channel comparison: Poisson modification is excluded by BOSS RSD data.

EFC’s coupling must enter through the Friedmann equation. The Poisson channel at $\alpha_{L2} = 0.34$ gives $\chi^2 = 18.2$ — excluded. The background channel gives $\chi^2 = 1.09$, mildly favoured over Λ CDM.

2.3 Physical Mechanism

Geometry: Comoving distances $D_M = \int dz'/H$ shrink slightly, pulling BAO predictions closer to BOSS data.

Growth: Enhanced Hubble friction in the growth ODE ($\delta + 2H\delta = 4\pi G \rho_m \delta$) suppresses late-time clustering, reducing $f\sigma_8$ toward observed values.

Both arise from the same single modification. No separate tuning is required.

3. Four-Channel Test

3.1 Channel 1: BAO Distances (BOSS DR12)

BOSS DR12 provides D_M/r_d and D_H/r_d at $z = 0.38, 0.51, 0.61$ (Alam et al. 2017).

z	D_M/r_d pull	D_H/r_d pull	Note
0.38	+0.29 σ	-1.10 σ	D_H mildly tense
0.51	-0.26 σ	+0.12 σ	Excellent
0.61	+0.14 σ	+0.74 σ	Good

Table 2. BAO distance pulls at $\beta = 0.08$.

$\chi^2 = 1.93$ (Λ CDM: 5.58). Mild preference, $\Delta\chi^2 = -3.65$.

3.2 Channel 2: Growth Rate (BOSS DR12 $f\sigma_8$)

z	$f\sigma_8$ (EFC)	$f\sigma_8$ (obs)	Pull	$T(z)$
0.38	0.464	0.497 ± 0.045	-0.73 σ	0.95
0.51	0.463	0.458 ± 0.038	+0.13 σ	0.99
0.61	0.460	0.436 ± 0.034	+0.71 σ	0.97

Table 3. Growth rate data vs EFC prediction at $\beta = 0.08$.

$\chi^2 = 1.09$ (Λ CDM: 1.34). Marginal preference, $\Delta\chi^2 = -0.25$. All pulls $< 1\sigma$.

3.3 Channel 3: CMB Lensing (ACT+SPT+Planck 2025)

Latest combined CMB lensing: $S_8^{\text{CMBL}} \equiv \sigma_8(\Omega_m/0.3)^{0.25} = 0.825 \pm 0.014$ (Qu et al. 2025, arXiv:2504.20038). The lensing kernel peaks at $z \sim 2$ where $T = 0.19$ — the modification is largely diluted.

Quantity	Λ CDM	EFC ($\beta = 0.08$)	Observed
$\sigma_{8,0}$	0.811	0.805	—
S_8 (CMB lensing)	0.821	0.815	0.825 ± 0.014
Pull	-0.28 σ	-0.71 σ	—

Table 4. CMB lensing comparison. Both models consistent; EFC slightly farther but $< 1\sigma$.

This channel is **neutral** — no kill, no improvement.

3.4 Channel 4: Type Ia Supernovae

z	$\Delta\mu$ (mag)	$\Delta D_L/D_L$	$T(z)$	Pantheon+ σ
0.1	-0.015	-0.7%	0.50	~ 0.04 mag
0.3	-0.029	-1.3%	0.94	~ 0.03 mag
0.5	-0.032	-1.5%	0.99	~ 0.03 mag
1.0	-0.028	-1.3%	0.69	~ 0.05 mag

Table 5. SN Ia distance modulus shifts. Maximum $\sim 1\sigma$; same direction as DESI dynamical-DE hint.

This channel is **neutral** — **consistent but not distinguishing**.

4. Combined Results

4.1 Four-Channel Summary

Channel	Status	Key metric	vs Λ CDM	Role
BAO distances	PASS	$\chi^2 = 1.93$	Mild pref. (−3.65)	Detection
Growth rate	PASS	$\chi^2 = 1.09$	Marginal (−0.25)	Detection
CMB lensing	PASS	pull = −0.71 σ	Slightly worse	Neutral
SN Ia	PASS	$\Delta\mu < 0.032$	Neutral	Neutral
Combined BAO+RSD		$\Delta\chi^2 = -3.89$	~2 σ pref.	—

Table 6. Four-channel summary. Same $\beta = 0.08$, same $T(a)$, zero kills.

Same $\beta = 0.08$, same $T(a)$, four independent channels, zero kills.

4.2 The σ_8 Landscape

Probe	σ_8	$\pm\sigma$	EFC pull	Regime	z
Planck 2018 primary	0.811	0.006	−1.0 σ	CMB	1100
Planck PR4 primary	0.807	0.006	−0.33 σ	CMB	1100
CMB lensing combined	0.811	0.019	−0.32 σ	$z \sim 2$	0.5–5
BOSS RSD (derived)	0.805	0.020	+0.00 σ	Late	~ 0.5
KiDS Legacy	0.748	0.022	+2.6 σ	Late	~ 0.5
DES Y3	0.765	0.023	+1.7 σ	Late	~ 0.5
EFC prediction	0.805	—	—	—	—

Table 7. σ_8 landscape. EFC sits between CMB and galaxy lensing values.

EFC reduces the S_8 tension from ~3 σ to ~2 σ but does not fully resolve it. Planck PR4 revises σ_8 to ~0.807, bringing EFC within 0.33 σ .

5. Regime Consistency: Why Three Passes Are One Consequence

Because $T(a) \propto (1 - \Omega_m(z))$, the modification amplitude is controlled by the dark-energy fraction at each epoch:

Epoch	z	$T(z)$	$ \mu - 1 $	Observational margin
BOSS (measured)	0.51	0.990	0.340	Detection channel
Ly α P1D (null)	3.0	0.107	3.7×10^{-2}	~10 $\times$ suppressed
CMB recombination	1100	9.7×10^{-11}	3.3×10^{-11}	> 10 $^8\times$ below bound
BBN	~10 8	< 10 $^{-30}$	< 10 $^{-25}$	> 10 $^{23}\times$ below bound

Table 8. Regime hierarchy from a single coupling law.

The genuine empirical content is that the same β produces a mild preference in two independent observables (BAO distances and growth rates) simultaneously at $z \sim 0.5$, without triggering kills in the other two channels.

6. Falsification Criteria

#	Observation	What it would show	Type
F1	Large $\mu - 1$ at $z > 2$ where $\Omega_m \approx 1$	Coupling not epoch-governed	Hard kill
F2	Growth enhancement at $z < 1$	Background channel wrong	Hard kill
F3	Poisson $\mu > 1$ detected at α_{L2} level	Architecture violated	Hard kill
F4	BAO–growth β split $> 3\sigma$	Single-channel fails	Hard kill
F5	CMB lensing $\sigma_8 > 3\sigma$ from EFC	Background inconsistent	Hard kill

Table 9. Falsification criteria. Current status: none triggered. Tightest: CMB lensing at -0.71σ .

7. Calibration and Interpretation Boundaries

All results are computed within a Λ CDM-calibrated framework. $T(z)$ uses $\Omega_m(z)$ from Planck 2018 parameters. The test verifies *internal consistency of EFC’s coupling structure given standard background calibration*, not a fully self-consistent EFC cosmology.

8. Discussion

The result is not a detection of new physics. It is a consistency test: one parameter ($\beta = 0.08$), one equation (modified Friedmann), one transition function, four probes, zero kills, mild preference in two.

The architectural constraint — background not Poisson — is perhaps the most important finding. It pins EFC to a specific mechanism: modified expansion producing indirect growth suppression via Hubble friction. EFC predicts slightly brighter supernovae at $z \sim 0.5$, in the same direction as the DESI dynamical-DE hint — testable with DESI DR2.

9. Conclusions

1. Poisson-equation coupling ($\mu > 1$) is excluded by BOSS RSD. EFC enters through the Friedmann equation.
2. $\beta = 0.08$ shows mild preference in combined BAO + RSD ($\Delta\chi^2 = -3.89, \sim 2\sigma$).
3. Independently consistent with CMB lensing (-0.71σ) and SN Ia ($\sim 1\sigma$). Four channels, zero kills.
4. Regime passes at $z = 3$ and $z = 1100$ are automatic consequences of $T(a) \rightarrow 0$ when $\Omega_m \rightarrow 1$.
5. Five falsification criteria (F1–F5) are provided. None currently triggered.

This is a consistency test, not a full MCMC analysis with covariances. Primary CMB anisotropies require a modified Boltzmann solver and are not tested here.

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